

COMPANION TO CDA V1.1

K8s Event Triage Platform

Project Status — Implementation Snapshot

Owner

Platform Engineering Team

Version

1.0 — Draft

Date

2026-04-28

Companion to

CDA v1.1

AGENDA

What we'll cover

#	SECTION	CONTENTS
01	Executive Snapshot	State of every component
02	System Design	Components · topology · routing · skills · HITL
03	Component Status	HITL · A2A · LGTM · agentgateway · skills · memory
04	Architecture Decisions	14 key calls + rationale
05	Agent Cards	~20 agents · 5 grouped patterns
06	Open Work	Pending items + owners

SECTION 01

Executive Snapshot

Where every moving part is today

STATE OF EVERY COMPONENT (1/2)

What's already in production

THEME	STATE	DETAIL
Event ingestion (Alloy → Event Hub → Argo Events)	PRODUCTION	Three-tier pipeline tested e2e on <code>proxmox-k8s</code>
Argo Workflows triage DAG	PRODUCTION	parse-otlp → fan-out → A2A → GitLab + Teams
KAgent A2A protocol	PRODUCTION	JSON-RPC 2.0 · <code>message/send</code> · agent-as-tool proven
agentgateway (replaced LiteLLM)	PRODUCTION	UAMI to Azure OpenAI · OTel native · MCP & A2A gateway
HITL Teams bot PoC	POC DONE	End-to-end proven · Istio VS+AuthZ · <code>smoke-test.sh</code>
GitLab ticketing	PRODUCTION	Direct REST today · MCP migration on roadmap
Specialist agent roster	PHASE 2	11 namespace agents deployed · ~20 planned
Skills (kagent 0.8.0+)	SPARSE	Mechanism deployed · library grows via learning loop

STATE OF EVERY COMPONENT (2/2)

What's designed · in flight · future

THEME	STATE	DETAIL
Memory (pgvector backend)	IN FLIGHT	pgvector deployed · <code>load_memory</code> wiring per-agent
Full A2A → HITL demo	IN FLIGHT	Joining Suspend → Adaptive Card → callback to specialist agents
Round-robin GitLab assignment	DESIGNED	Needs dedicated GitLab project + roster ConfigMap
Managed LGTM integration	DESIGNED	<code>managed-lgtm-integration/</code> · pending platform-team Q&A
GitLab MCP integration	FUTURE	Sequenced after HITL demo
ServiceNow / Azure DevOps MCP	FUTURE	After GitLab MCP
<code>gitops-remediation-agent</code> build	FUTURE	After GitLab MCP
Real Teams Bot (replace Logic App)	FUTURE	App registration + Bot Framework

HEADLINE NUMBERS

The shape of the thing

METRIC	VALUE	WHAT IT MEANS
Agents	~20	Specialist agents · 11 deployed · 9 planned
Triage time	~60s	Event → diagnosis · vs. 30–60 min manual
Tiers	3	critical · warnings · infra · per consumer group
Dedup layers	3	Alloy · Sensor · Script TTL
Notification surface	2	GitLab issues · Teams Adaptive Cards
Cluster coverage	N+1	One management cluster · N worker clusters

Goal state: zero unresolved events — every K8s warning either auto-resolved by an agent or round-robin assigned to a human within minutes.

SECTION 02

System Design

Topology · routing · skills · output

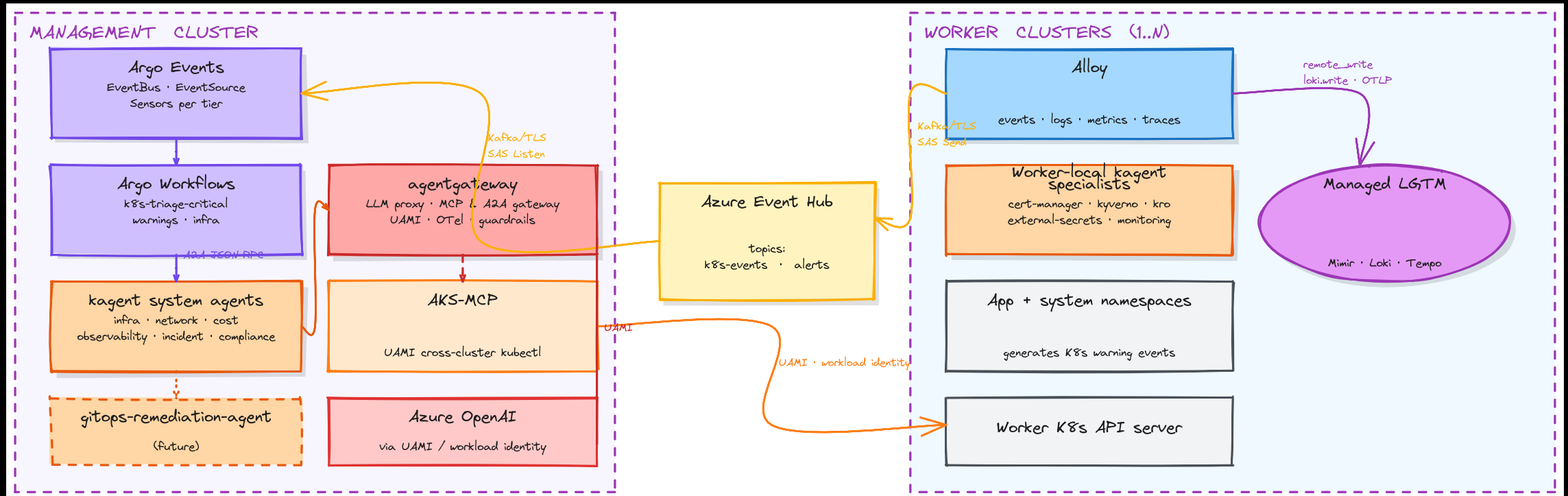
WHAT WE'RE BUILDING

AI-driven triage for Kubernetes warning events

A management cluster watches **N worker clusters**. When a K8s warning event fires, it flows through Argo Events → Argo Workflows, gets dispatched to a **specialist kagent agent** that diagnoses the root cause via LLM, and the result becomes a **GitLab issue + Teams Adaptive Card**. Write actions are gated by a **Human-in-the-Loop** approval before any cluster change.

COMPONENT	ROLE
Alloy	Per-cluster log/metric/event collector — forwards K8s warnings to Event Hub
Azure Event Hub	Cross-cluster event bus — Kafka-protocol, SAS-authenticated
Argo Events	EventBus (NATS) · EventSource (Kafka consumer) · per-tier Sensors
Argo Workflows	Triage DAG — <code>parse-otlp</code> → fan-out → A2A → output
kagent	Specialist agent runtime — system prompt + skills + memory + tools
agentgateway	LLM proxy + MCP & A2A gateway · UAMI to Azure OpenAI · OTel native
AKS-MCP	Cross-cluster <code>kubectl</code> for the agents · UAMI / workload identity
HITL gate	Teams Adaptive Card via Logic App · Istio HMAC-verified callback
Platform Engineering · Project Status v1.0 · 2026-04-28	
GitLab + Teams	Output surfaces — ticket destination + notifications

Brain in the centre, eyes everywhere



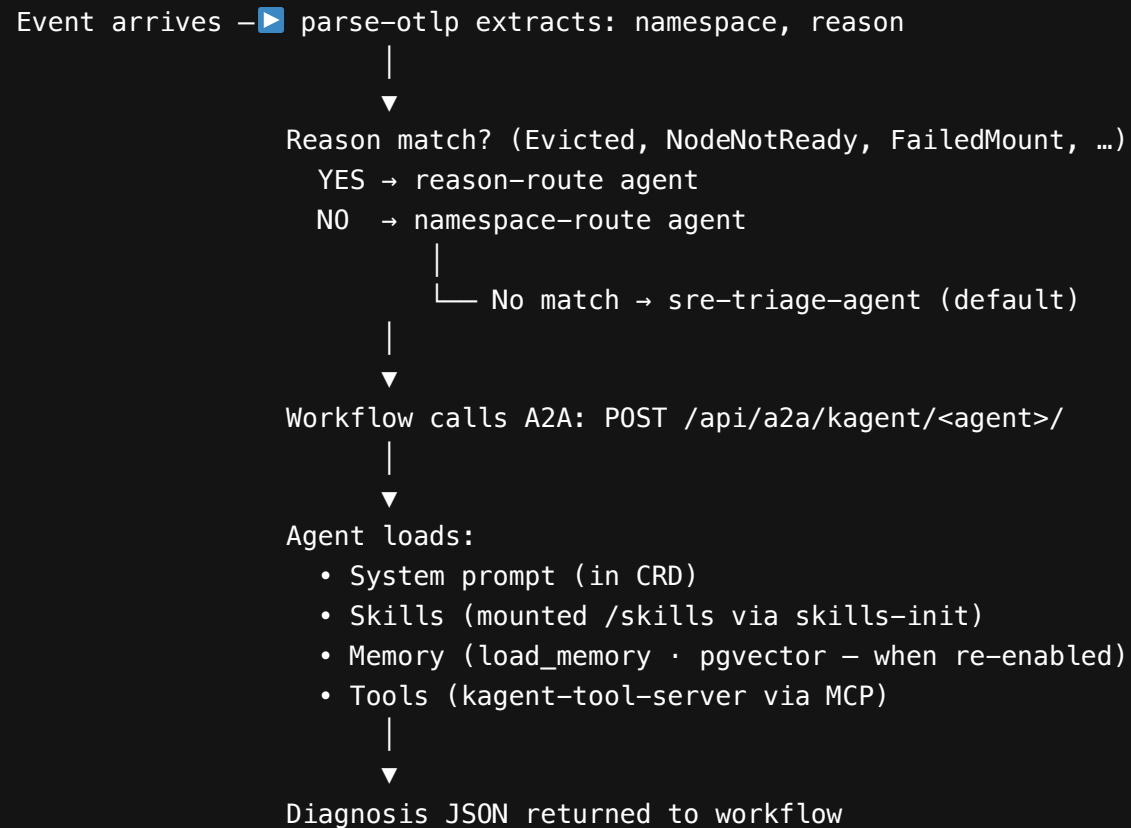
WHY SPLIT MGMT VS WORKER

Different concerns, different blast radius

CONCERN	MANAGEMENT CLUSTER	WORKER CLUSTER
Cross-cluster orchestration	Yes — AKS-MCP, multi-cluster routing	No — only sees self
System namespaces of interest	argo, argo-events, kagent, agentgateway-system, aks-mcp	cert-manager, kyverno, kro, external-secrets, monitoring, istio-system, flux-system
Triage location	Aggregates events from N workers	Local triage when EventHub round-trip is unnecessary
LLM access	Yes (via agentgateway)	Yes (via agentgateway, mgmt or local)
Latency	Slower — round-trip via Event Hub	Faster — local A2A

Worker-local triage shortens loops from seconds to milliseconds for namespaces that don't need cross-cluster context (cert-manager, kyverno, kro etc.).

Two ConfigMaps drive the dispatch



Routing config: `eventhub-otlp-pipeline/tier-critical/agent-routing.yaml`

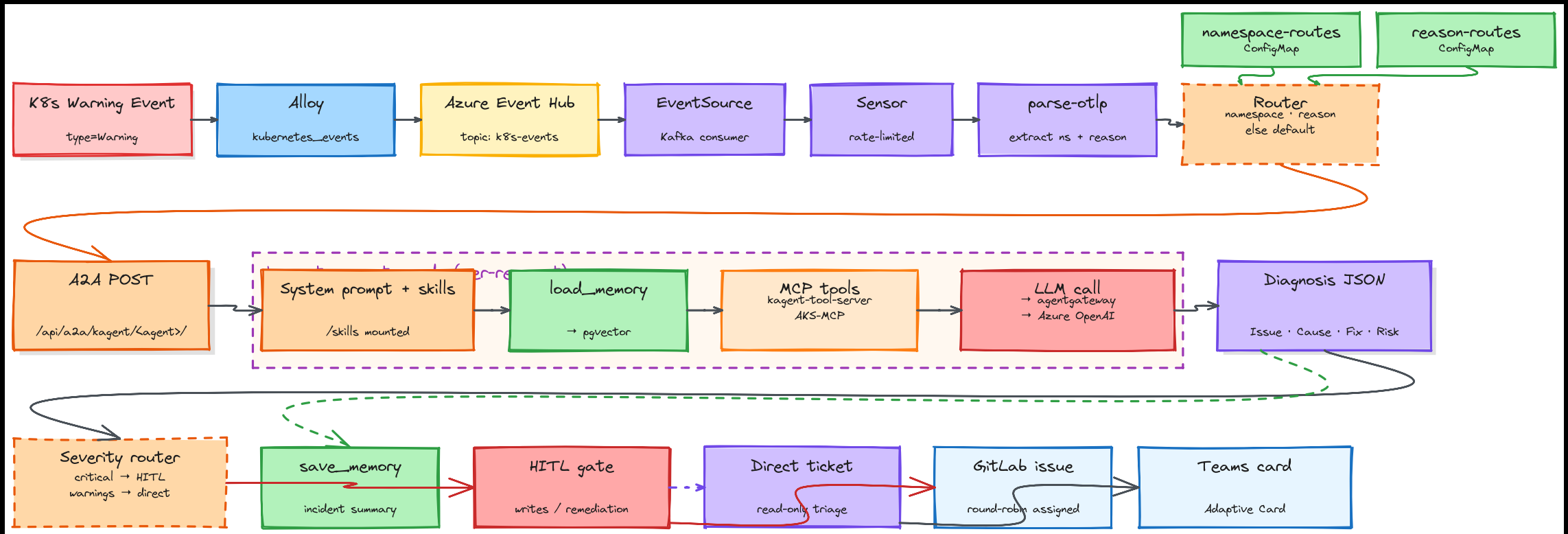
How agents pick up domain knowledge

LAYER	SOURCE	LOADED WHEN	MECHANISM
System prompt	<code>Agent.spec.declarative.systemMessage</code>	Pod start	Inline in CRD
A2A skill metadata	<code>a2aConfig.skills</code>	Pod start	Discovery / catalog only — not executed
Executable skills	<code>spec.skills.gitRefs</code> or OCI	Pod start	<code>skills-init</code> initContainer clones to <code>emptyDir</code> at <code>/skills</code>
Tools	<code>spec.declarative.tools</code> (MCP)	Per-request	RemoteMCPServer or Service · agentgateway can be central MCP gateway
Memory	<code>spec.declarative.memory</code> (when re-enabled)	Per-request	<code>load_memory(query)</code> · <code>save_memory(text)</code> · pgvector
Prompt template data	<code>promptTemplate.dataSources</code>	Per-request	ConfigMap-backed builtins + per-namespace overrides

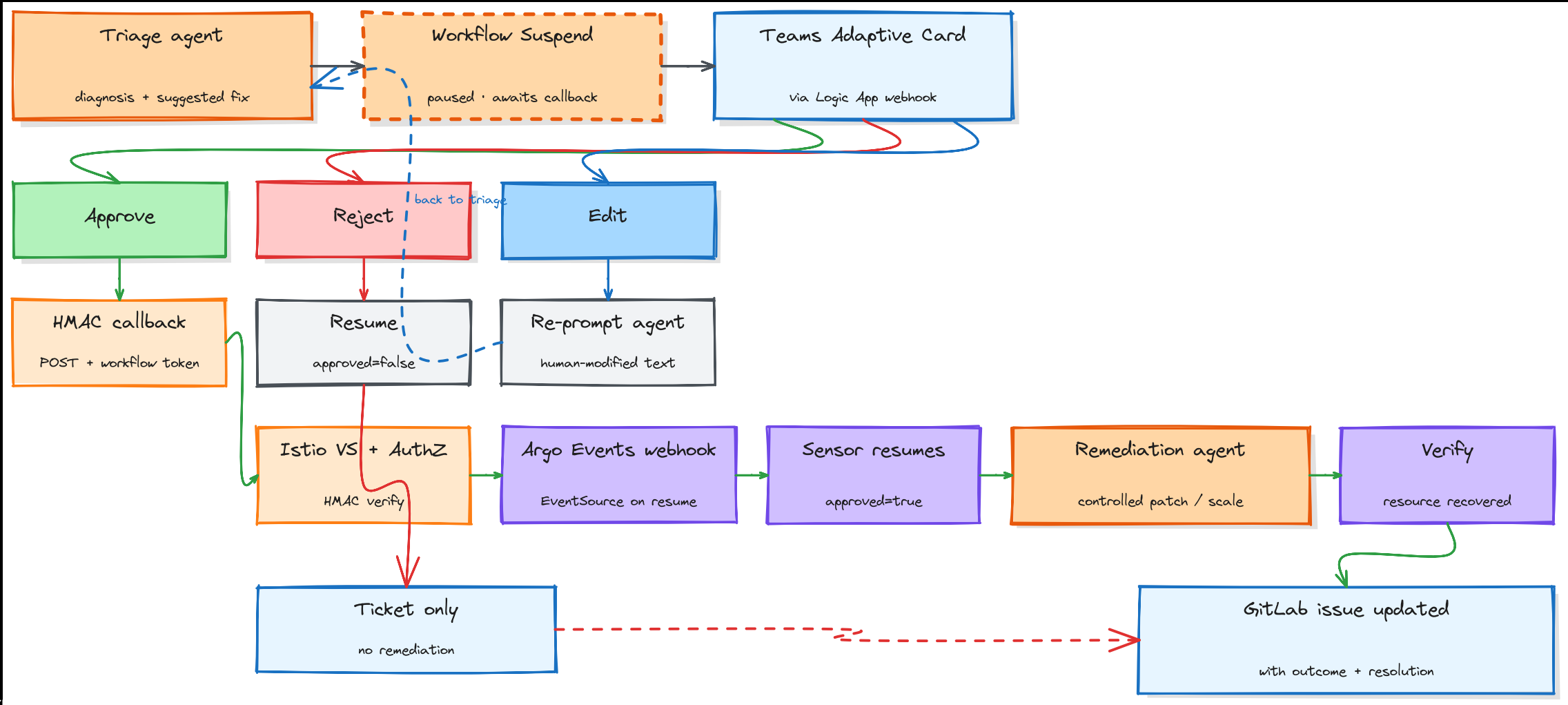
Critical naming rule: `spec.skills` (executable bundles) ≠ `a2aConfig.skills` (metadata only).

PER-REQUEST TRIAGE FLOW

Event → routing → agent → diagnosis → output



Approve · Reject · Edit — gate before any write



TICKETING INTEGRATIONS

Where triage lands

INTEGRATION	STATUS	MECHANISM	USE CASE
GitLab (REST API today)	PRODUCTION	Direct API · workflow step	Default ticket destination · round-robin assignment
Microsoft Teams	PRODUCTION	Logic App webhook	Adaptive Card notifications · HITL surface
GitLab MCP	FUTURE	gitlab-mcp-server	Replace REST calls with MCP for richer ticket interactions
ServiceNow MCP	FUTURE	servicenow-mcp	Enterprise incident · severity mapping · escalation chains
Azure DevOps MCP	FUTURE	ado-mcp	Work item creation in Boards

All three MCPs sequenced **after** the full A2A → HITL demo lands. The **incident-agent** will select target(s) by severity + namespace ownership when MCPs are wired.

SECTION 03

Component Status

Six pillars · where each one is

Workflow + Teams PoC done • A2A integration in flight

ELEMENT	STATUS
Teams Adaptive Card payload via Logic App	PRODUCTION
Argo Events webhook EventSource (callback)	PRODUCTION
Istio VirtualService + AuthorizationPolicy (HMAC)	PRODUCTION
Workflow Suspend → Adaptive Card → resume	POC DONE
Mock Teams bot (stand-in until real bot lands)	POC DONE
Full A2A → HITL demo (specialist agents)	IN FLIGHT
Real Teams bot (replacing Logic App)	FUTURE

Current focus: join the proven gate to specialist-agent A2A so a triage agent's suggested remediation can suspend, surface in Teams, and resume on approval.

JSON-RPC 2.0 with sharp edges

```
POST /api/a2a/kagent/<agent>/

{ "jsonrpc": "2.0", "id": "...", "method": "message/send",
  "params": { "message": { "role": "user",
    "parts": [{"kind":"text","text":"..."}] } } }
```

COMMON GOTCHA	SYMPTOM
Missing trailing slash on URL	404 silent
Method <code>tasks/send</code> (old docs)	"method not supported"
<code>parts</code> missing <code>"kind": "text"</code>	parse error
Session API used for sending	403 even with right user

Agent-as-tool pattern: kagent supports `tools: [{type: Agent, agent: {name, namespace}}]` natively. Proven March 2026 with the Kimi coordinator (6/6 consecutive tool calls including a revision loop). Same path: `cert-manager-agent` → `gitops-remediation-agent`.

MANAGED LGTM INTEGRATION

Alloy is the only handle — bidirectional

DIRECTION	PATTERN	COMPONENT
Push metrics	ServiceMonitor / PodMonitor → remote_write	<code>prometheus.operator.*</code> → <code>prometheus.remote_write</code>
Push logs	Pod logs + K8s events	<code>loki.source.kubernetes</code> + <code>loki.source.kubernetes_events</code>
Push traces	OTLP receiver → Tempo	<code>otelcol.receiver.otlp</code> → <code>otelcol.exporter.otlp</code>
Provision rules	PrometheusRule + LokiRule CRDs → managed Ruler	<code>mimir.rules.kubernetes</code> + <code>loki.rules.kubernetes</code>
Loop alerts back	AM webhook → Kafka → Event Hub	<code>loki.source.api</code> → <code>otelcol.exporter.kafka</code>
Agent anomalies	Recording-rule baselines + z-score / cohort	LogQL loop detection
Triage prompt enrichment	Workflow queries Mimir/Loki at triage-time	Splices context into KAgent prompt

DESIGNED

· Pending answers from platform team on Q1–Q14 · ``aks-mgmt-stack/k8s-event-triage/managed-lgtm-integration/``

AGENTGATEWAY

LiteLLM replaced • agentgateway is current state

✓ **Migration complete.** All agents route LLM, MCP, and A2A traffic through agentgateway. LiteLLM removed. PostgreSQL retained for kagent memory.

WHY WE MOVED

UAMI / workload identity to Azure OpenAI — no keys to rotate

OTel native — `agentgateway_gen_ai_*` token + latency metrics

MCP gateway for centralised auth + federation

A2A gateway for inter-agent routing + auth

Per-route budgets + guardrails (regex / moderation / custom webhooks)

WHAT STAYED

kagent Agent CRDs — only ModelConfig `baseUrl` changed

A2A protocol — same JSON-RPC 2.0 contract

Argo Workflows DAGs — no change at all

PostgreSQL — retained for kagent memory

SKILLS MECHANISM

Git-versioned, mounted via initContainer

```
Agent CRD spec.skills.gitRefs:  
- url: https://github.com/your-org/your-skills-repo.git  
  ref: main  
  path: skills/cert-diagnostics  
  name: cert-diagnostics  
  |  
  ▼  
  kagent injects "skills-init" initContainer  
  |  
  ▼  
  Clones repos to emptyDir at /skills  
  |  
  ▼  
  Main agent container mounts /skills read-only  
  Discovers skills via SKILL.md frontmatter  
  Executes scripts via built-in BashTool (sandboxed)
```

STATUS

Mechanism deployed and proven · library currently shallow

GROWTH

Driven by the **learning loop** — every triaged ticket closes only when an agent skill is updated

MEMORY STATUS

Designed · backend ready · integration in flight

In flight — pgvector deployed · `load_memory` / `save_memory` designed · wiring into specialist agents alongside the A2A → HITL demo.

ELEMENT	STATUS
PostgreSQL + pgvector backend	DEPLOYED
Embedding ModelConfig	DESIGNED · separate from inference
<code>load_memory</code> / <code>save_memory</code> tools	WIRING per-agent
Memory + A2A design doc	DONE
Cross-incident pattern matching	DESIGNED · keys: <code>event_reason</code> + <code>namespace</code>
pgvectorscale upgrade path	OPTIONAL for StreamingDiskANN

Goal: agents check past incidents before re-investigating · "Seen before — incident #142, ACME fix worked" · 90-day TTL.

SECTION 04

Architecture Decisions

14 calls that shaped the design

DECISIONS 1 - 7

Specialisation, safety, contracts

#	DECISION	RATIONALE
1	Specialist agents per namespace, not generalist	Smaller models hallucinate when omniscient · focused prompts win
2	Read-only by default · write only via promotion	Blast radius is the hard control · RBAC-enforced not prompt-enforced
3	Namespace anchoring (CRITICAL: use exact namespace "X")	Empirical fix for Qwen 14B typos · belt-and-braces alongside Kyverno
4	A2A as the inter-agent contract	Standardised JSON-RPC · kagent native · agentgateway-routable
5	No kubectl edit from agents — patch / apply only	Interactive editors via MCP break · patch is deterministic + auditable
6	GitOps remediation via MR, not direct cluster writes	Agent never kubectl apply s · opens MR · Flux applies · auditable, reversible
7	Triage agent ≠ remediation agent	Two CRDs, two SAs, two RBAC levels · never an upgrade in place

DECISIONS 8 – 14

Operations, gates, scaling

#	DECISION	RATIONALE
8	HITL gate before any write action	Approve/Reject/Edit · auto-graduates to no-gate after N successes
9	Worker-local triage where possible	Eliminates Event Hub round-trip for self-contained namespaces
10	Round-robin GitLab assignment, no namespace ownership	Builds breadth · visible at standups via Kanban
11	Skills are git-versioned, not in-line in CRDs	<code>gitRefs</code> · skills library reviewable in PRs · CRD stays small
12	Agent prompts committed to Git	<code>systemMessage</code> is a code artefact · changes go through PR review
13	3-layer dedup (Alloy / Sensor rate / script TTL)	Single layer brittle under storms · three layers means one can fail
14	agentgateway as single LLM exit point	One place for token tracking, guardrails, UAMI auth, OTel · agents don't know about Azure OpenAI directly

SECTION 05

Agent Cards

~20 agents · 5 grouped patterns

COMMON CARD FIELDS

Shared across all cards (unless flagged)

FIELD	COMMON VALUE
Process	Receive A2A <code>message/send</code> → load skills + memory → call MCP tools → summarise → return JSON-RPC result. Optional A2A call to <code>gitops-remediation-agent</code> for fixes.
Communication	Inbound: A2A from Argo Workflow. Outbound: A2A to other agents · MCP to <code>kagent-tool-server</code> , <code>aks-mcp</code> , <code>gitlab-mcp-server</code> , <code>git-mcp-server</code> .
Tools (read)	<code>k8s_get_resources</code> · <code>k8s_describe_resource</code> · <code>k8s_get_pod_logs</code> · <code>k8s_get_events</code> · <code>k8s_get_resource_yaml</code>
Tools (write — remediation tier only)	<code>k8s_patch_resource</code> · <code>k8s_apply_manifest</code> · <code>k8s_delete_resource</code> · <code>k8s_label_resource</code> · <code>k8s_annotate_resource</code> · <code>k8s_execute_command</code>
Knowledge	System prompt + <code>/skills</code> git bundle + (when enabled) pgvector memory + <code>kagent-builtin-prompts</code> ConfigMap
Evaluation	Output schema enforced (Issue/Affected/Root Cause/Remediation/Risk/Verification) · workflow grades fields and rejects malformed
Security	Read-only RBAC default · write via explicit RoleBinding to specific namespaces · UAMI scope per cluster · prompt constraints
Model routing	All → agentgateway → Azure OpenAI (UAMI) or local vLLM · same <code>default-model-config</code>

CARD A — TRIAGE SPECIALIST

The dominant pattern (covers ~20 agents)

PROCESS

Reactive — invoked by workflow on K8s Warning event. Optionally proactive via CronWorkflow (compliance, cost, security).

USER PERSONA

SRE / Platform Engineer responding to a triage notification — wants structured diagnosis + ranked remediation + exact kubectl commands.

PROBLEM

Domain-specific K8s failures hit a namespace · on-call lacks deep specialist knowledge · agent collapses 30–60 min into seconds.

ROLE

Read-only investigator + remediation suggester. Writes nothing. Returns analysis JSON to the workflow.

Applies to: cert-manager · kyverno · kro · external-secrets · reloader · flux-system · gatekeeper-system · istio-ingress · istio-system · kube-system · dns · database · observability · cost · change · network · storage · infra · security · compliance.

CARD A — SPECIALIST SKILLS

What lives in `/skills/<agent>/`

AGENT	SKILLS BUNDLE
<code>cert-manager-agent</code>	<code>cert-diagnostics</code> · ACME challenge inspection · issuer status · expiry scans
<code>network-agent</code>	<code>cni-health-probe</code> · <code>ingress-backend-walker</code> · <code>istio-config-analyse</code>
<code>storage-agent</code>	<code>pvc-binding-trace</code> · <code>longhorn-volume-health</code> · <code>csi-driver-probe</code>
<code>infra-agent</code>	<code>node-condition-decoder</code> · <code>kubelet-log-grep</code> · <code>inotify-pressure-check</code>
<code>security-agent</code>	<code>pss-violation-decoder</code> · <code>falco-event-correlator</code> · <code>rbac-audit</code>
<code>cost-agent</code>	<code>right-size-recommender</code> · <code>idle-resource-finder</code> · <code>node-utilisation-report</code>
<code>compliance-agent</code>	<code>label-compliance-scan</code> · <code>quota-coverage-scan</code> · <code>network-policy-coverage</code>
<code>observability-agent</code>	<code>prometheus-target-health</code> · <code>loki-stream-audit</code> · <code>cardinality-explorer</code>

Full per-agent definitions: `aks-mgmt-stack/k8s-event-triage/AGENT-ROSTER.md`

CARD B — REMEDIATION AGENT

Limited write · gated by HITL approval

PROCESS

Invoked only after HITL approval. Receives diagnosis + approval token. Executes recommended kubectl actions in a controlled subset.

ALLOWED ACTIONS

patch · scale · restart · label/annotate

FORBIDDEN

Delete CRDs · delete PVCs with data · restart all replicas at once · anything outside approved namespace

SECURITY

Approval token verified against workflow HMAC · replay protection · action whitelist enforced server-side · failures escalate, never retry

Applies to: sre-remediation-agent. All writes go through GitOps where possible · direct cluster writes only for break-glass.

Future · branch + commit + MR · never `kubectl` `apply`

Future work — sequenced after the A2A → HITL demo lands

Concept proven via the Kimi 6/6 A2A pattern · build starts once GitLab MCP integration begins.

PROCESS

Receives A2A call from triage agent describing needed config change. Clones GitOps repo · branch · diff · commit · MR. Never pushes to main.

USER PERSONA

SRE reviewing the MR — wants clean single-purpose diff with agent diagnosis as MR description and GitLab triage issue linked.

TOOLS NEEDED

`git-mcp-server` (clone/checkout/commit/push/diff/file_*) · `gitlab-mcp-server` with `requireApproval` on `create_merge_request`

SAFETY CONTROLS (PLANNED)

Bot account scoped to single GitOps project · MR title `[Auto-Fix]` · one change per MR · GPG-signed · detect-secrets pre-commit.

CARD D — SYSTEM AGENTS (MGMT CLUSTER)

Cluster-wide read · scheduled CronWorkflow

AGENT	SPECIALIST SKILLS	MOSTLY
incident-agent	incident-correlator · severity-escalator · mttr-calculator	Process management
change-agent	rollout-watcher · canary-comparator · argocd-sync-checker	Deployment health
compliance-agent	weekly-compliance-report · audit-evidence-exporter	Reports + scorecards
cost-agent	weekly-cost-report · right-sizing-recommender	FinOps reports
observability-agent	monitoring-stack-health · cardinality-explosion-detector	Stack-of-stacks monitoring

User persona: Engineering leadership · FinOps · SRE management. Outputs: weekly reports as GitLab issues · weekly Teams summaries · ServiceNow / Azure DevOps tickets when those MCPs land. RBAC: read across all namespaces · incident-agent has write access to ticketing systems only.

Cluster-readiness certifier · agent-graded verdict

Reuses `sre-triage-agent` · invoked from the `uk8s-certification-v2` RGD. Argo Workflow runs validation tasks → ConfigMap aggregation → A2A handoff → strict-schema verdict back into Argo `outputs.parameters` .

ASPECT	DETAIL
Process	Per-workflow ConfigMap captures pass/fail counts · <code>aggregate-certification</code> sums to CERTIFIED / FAILED · <code>agent-triage</code> sends report over A2A · agent returns <code>{severity, blast_radius, failed_checks[], first_remediation}</code>
User persona	Cluster operator certifying a new or updated cluster — wants a single gate plus an AI-graded list of what to fix first
Outputs	<code>severity</code> (CRITICAL / WARNING / INFO / AGENT_UNAVAILABLE / PARSE_ERROR) and <code>verdict</code> as Argo output params — branchable by Teams / exit hook / GitLab issue
Optional gating	<code>failOnCritical</code> (off by default) fails the workflow when severity = CRITICAL — upstream alerting fires
Source	<code>infra-stack/kro-stack/definitions/uk8s-certification-v2.yaml</code> · doc: <code>UK8S-CERTIFICATION-V2.md</code>

SECTION 06

Open Work

Pending items + ownership

PENDING ITEMS (1/2)

In flight + designed

ITEM	STATUS	OWNER	BLOCKER
Full A2A → HITL demo	IN FLIGHT	Engineering	Wiring specialist agents into proven HITL gate
Memory integration into specialist agents	IN FLIGHT	Engineering	load_memory / save_memory per-agent rollout
Specialist agents on workers	11 OF ~20	Engineering → SRE	Phase 3 of ENGINEERING-ROLLOUT.md
Skills library expansion	CONTINUOUS	All engineers	Driven by learning loop
LGTM integration deployment	DESIGNED	Engineering + Platform	OPEN-QUESTIONS Q1-Q14
Round-robin GitLab assignment	DESIGNED	Engineering	Needs dedicated GitLab project + roster ConfigMap

PENDING ITEMS (2/2)

Future · sequenced after HITL demo

ITEM		STATUS	OWNER	BLOCKER
GitLab MCP integration		FUTURE	Engineering	After HITL demo · replaces direct REST calls
<code>gitops-remediation-agent</code> build		FUTURE	Engineering	After GitLab MCP lands
ServiceNow MCP integration		FUTURE	TBD	After GitLab MCP
Azure DevOps MCP integration		FUTURE	TBD	After ServiceNow MCP
Real Teams Bot (replace Logic App)		FUTURE	TBD	App registration + Bot Framework deploy

DOCUMENT MAP

Where to go for the detail

DOCUMENT	PURPOSE
<code>docs/CDA-DESIGN-AUTHORITY.md</code> (v1.1)	Formal design authority position
<code>docs/PROJECT-STATUS.md</code>	Long-form companion to this deck
<code>docs/CDA-SLIDES.md</code> (v1.1)	CDA as slides
<code>docs/SAD-THREAT-MODEL.md</code>	STRIDE threat model
<code>docs/SAD-LOGGING-MONITORING-AUTH.md</code>	Logging/monitoring/auth
<code>docs/SAD-LOGGING-MONITORING-LLM.md</code>	LLM governance
<code>worker-cluster-bundle/SKILLS-AND-REMEDIATION.md</code>	Skills loading mechanism
<code>worker-cluster-bundle/MEMORY-AND-A2A-REMEDIATION-DESIGN.md</code>	Memory + A2A remediation design
<code>worker-cluster-bundle/AGENTGATEWAY-TRANSITION.md</code>	LiteLLM → agentgateway migration
<code>aks-mgmt-stack/k8s-event-triage/managed-lgtm-integration/</code>	Managed LGTM design + agent anomalies
<code>aks-mgmt-stack/k8s-event-triage/AGENT-ROSTER.md</code>	Full per-agent specifications

THANK YOU

Questions?

Platform Engineering Team

Companion document: `docs/PROJECT-STATUS.md`

Render: `marp PROJECT-STATUS-SLIDES.md --pdf --allow-local-files`